ENGINE TECHNICAL DATA

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Item	Specification
Maximum distortion, head gasket side of the cylinder head	0.06 mm {0.002 in}
Maximum distortion of intake manifold and turbocharger installation surface	Intake manifold installation surface: 0.05 mm {0.002 in} Turbocharger installation surface: 0.05 mm {0.002 in}
Standard valve seat contact width	1.2—2.0 mm {0.048—0.078 in}
Valve seat angle	45°
Standard valve seat sinkage amount	IN: 0.39—0.79 mm {0.016—0.031 in} EX: 0.47—0.87 mm {0.019—0.034 in}
Standard valve head margin thickness	IN: 1.50—1.70 mm {0.0591—0.0669 in} EX: 1.68—1.88 mm {0.0662—0.0740 in}
Standard valve length	IN: 97.07—97.37 mm {3.822—3.833 in} EX: 96.99—97.29 mm {3.819—3.830 in}
Minimum valve length	IN: 96.85 mm {3.813 in} EX: 96.77 mm {3.810 in}
Standard valve stem diameter	IN: 5.470—5.485 mm {0.2154—0.2159 in} EX: 5.465—5.480 mm {0.2152—0.2157 in}
Minimum valve stem diameter	IN: 5.424 mm {0.2135 in} EX: 5.419 mm {0.2133 in}
Standard valve guide inner diameter	IN: 5.510—5.530 mm {0.2170—0.2177 in} EX: 5.510—5.530 mm {0.2170—0.2177 in}
Standard clearance between valve stem and guide	IN: 0.025—0.060 mm {0.0010—0.0023 in} EX: 0.030—0.065 mm {0.0012—0.0025 in}
Maximum clearance between valve stem and guide	0.10 mm {0.0039 in}
Standard valve guide projection height	14.8—15.3 mm {0.583—0.602 in}
Valve spring installation height	IN: When pressurized with spring force of 128.3—141.7 N {13.09—14.44 kgf, 28.85—31.85 lbf}, spring height is 35.0 mm {1.38 in} EX (TYPE A): When pressurized with spring force of 137.8—152.2 N {14.06—15.52 kgf, 30.98—34.21 lbf}, spring height is 35.0 mm {1.38 in} EX (TYPE B): When pressurized with spring force of 209—231 N {21.4—23.5 kgf, 47.0—51.9 lbf}, spring height is 35.0 mm {1.38 in}
Maximum valve spring off-square	IN: 2.0 ° (1.5 mm {0.059 in}) EX (TYPE A): 2.0 ° (1.7 mm {0.067 in}) EX (TYPE B): 2.0 ° (1.9 mm {0.075 in})
Maximum camshaft runout	0.030 mm {0.0012 in}
Standard cam height [intake camshaft]	34.12 mm {1.343 in}
Standard cam height [exhaust camshaft]	34.48 mm {1.357 in}
Minimum cam height [intake camshaft]	34.05 mm {1.341 in}
Minimum cam height [exhaust camshaft]	34.41 mm {1.355 in}
Standard camshaft journal diameter [intake camshaft]	No.1 journal: 33.950—33.970 mm {1.3367—1.3374 in} Except No.1 journal: 25.950—25.975 mm {1.0217—1.0226 in}
Standard camshaft journal diameter [exhaust camshaft]	No.1 journal: 33.950—33.975 mm {1.3367—1.3375 in} Except No.1 journal: 25.950—25.975 mm {1.0217—1.0226 in}
Minimum camshaft journal diameter	No.1 journal: 33.920 mm {1.3354 in} Except No.1 journal: 25.920 mm {1.0205 in}
Standard camshaft journal oil clearance [intake camshaft]	No.1 journal: 0.030—0.075 mm {0.0012—0.0029 in} Except No.1 journal: 0.025—0.071 mm {0.0010—0.0027 in}
Standard camshaft journal oil clearance [exhaust camshaft]	No.1 journal: 0.025—0.075 mm {0.0010—0.0029 in} Except No.1 journal: 0.025—0.071 mm {0.0010—0.0027 in}
Maximum camshaft journal oil clearance	No.1 journal: 0.105 mm {0.00413 in} Except No.1 journal: 0.101 mm {0.00398 in}
Standard camshaft end play	0.07—0.22 mm {0.003—0.008 in}
Maximum camshaft end play	0.98 mm {0.039 in}
Maximum distortion, head gasket side of the cylinder block [Other than between cylinder bores]	0.10 mm {0.0039 in}
Maximum distortion, head gasket side of the cylinder block [between cylinder bores]	Distortion on head gasket surface is convex: 0.015 mm {0.00059 in} Distortion on head gasket surface is concave: 0.045 mm {0.0018 in}
Standard cylinder bore diameter	76.000—76.022 mm {2.9922—2.9929 in}
Oil jet valve opening pressure	70—110 kPa {0.72—1.12 kgf/cm^2, 11.0—15.9 psi}
Standard piston outer diameter	75.958—75.984 mm {2.9905—2.9914 in}

Item	Specification
Standard clearance between piston and cylinder	0.031—0.049 mm {0.0013—0.0019 in}
Maximum clearance between piston and cylinder	0.067 mm {0.0026 in}
Standard clearance between piston ring and ring groove	Top: 0.10—0.14 mm {0.0040—0.0055 in} Second: 0.05—0.09 mm {0.002—0.003 in} Oil: 0.05—0.09 mm {0.002—0.003 in}
Maximum clearance between piston ring and ring groove	Top: 0.16 mm {0.0063 in} Second: 0.12 mm {0.0047 in} Oil: 0.11 mm {0.0043 in}
Standard piston ring end gap	Top: 0.17—0.22 mm {0.0067—0.0086 in} Second: 0.17—0.27 mm {0.007—0.010 in} Oil (rail): 0.15—0.25 mm {0.0060—0.0098 in}
Maximum piston ring end gap	Top: 0.38 mm {0.015 in} Second: 0.49 mm {0.019 in} Oil (rail): 0.46 mm {0.018 in}
Standard piston pin outer diameter	23.995—24.000 mm {0.94469—0.94488 in}
Standard piston pin hole diameter	24.005—24.013 mm {0.94508—0.94539 in}
Standard clearance between piston pin hole diameter and piston pin outer diameter	0.005—0.018 mm {0.0002—0.0007 in}
Standard connecting rod small end inner diameter	24.010—24.021 mm {0.94528—0.94570 in}
Standard clearance between connecting rod small end inner diameter and piston pin outer diameter	0.010—0.026 mm {0.0004—0.0010 in}
Maximum connecting rod bending	0.08 mm {0.003 in}
Maximum connecting rod distortion	0.08 mm {0.003 in}
Connecting rod center-to-center distance	138.19 mm {5.4406 in}
Standard side clearance at the large end of connecting rod	0.14—0.36 mm {0.006—0.014 in}
Maximum side clearance at the large end of connecting rod	0.512 mm {0.0202 in}
Standard bearing oil clearance at the large end of the connecting rod	0.026—0.052 mm {0.0011—0.0020 in}
Maximum bearing oil clearance at the large end of the connecting rod	0.10 mm {0.00394 in}
Connecting rod bearing size	STD: 1.501—1.520 mm {0.05910—0.05984 in} OS 0.25: 1.626—1.633 mm {0.06402—0.06429 in} OS 0.50: 1.751—1.758 mm {0.06894—0.06921 in}
Standard crankshaft end play	0.08—0.29 mm {0.004—0.011 in}
Maximum crankshaft end play	0.30 mm {0.012 in}
Thrust bearing size	STD: 2.500—2.550 mm {0.0985—0.1003 in} OS 0.25: 2.625—2.675 mm {0.1034—0.1053 in}
Maximum main journal runout	0.10 mm {0.0039 in}
Standard main journal diameter	46.980—47.000 mm {1.8497—1.8503 in}
Maximum main journal off-round	0.005 mm {0.0002 in}
Standard crank pin diameter	46.980—47.000 mm {1.8497—1.8503 in}
Maximum crank pin off-round	0.005 mm {0.0002 in}
Standard main journal oil clearance	0.016—0.039 mm {0.0007—0.0015 in}
Maximum main journal oil clearance	0.084 mm {0.0033 in}
Main bearing size	STD: 2.489—2.510 mm {0.0980—0.0988 in} OS 0.25: 2.614—2.620 mm {0.1030—0.1031 in} OS 0.50: 2.739—2.745 mm {0.1079—0.1080 in}
Dual-mass flywheel guide pin projection maximum amount	11.0—12.0 mm {0.434—0.472 in}
Standard cylinder head bolt length	145.2—145.8 mm {5.717—5.740 in}
Maximum cylinder head bolt length	146.5 mm {5.768 in}
Standard connecting rod bolt length	41.7—42.3 mm {1.65—1.66 in}
Maximum connecting rod bolt length	42.7 mm {1.68 in}
Rear oil seal press-in amount	0—0.5 mm {0—0.019 in} from cylinder block
Front oil seal press-in amount	0—0.5 mm {0—0.019 in} from engine front cover

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